

RJ11 and RJ45

When you look at the rear of a router you'll see two about 5 rectangular sockets with 4 being larger than the fifth. The bigger ones are RJ45 and the smaller one is RJ11. Telephone networks use RJ11 and ethernet cabling uses RJ45.

The one on the left is the RJ11 socket while the 5 bigger ones on the right are the RJ45 sockets.

Cat5/Cat5e/Cat6/Cat6a

Ethernet cables follow a twisted pair design, i.e., two wires are twisted together and the twists/ meter is limited. There are 8 such wires and 4 twisted pairs formed using them. The way these are wired are also standardised. There is the 568A and the 568B standard which decides which wire gets paired with which other. It pretty much doesn't matter which convention you follow as long as it is the same at both ends. The twisting in the cable is of prime importance. Every cable emits some interference, twisting two cables cancels out the said interference and reduces loss of signal strength due to attenuation. The more the twists the greater frequency the cable can handle, subsequently the higher the frequency the greater the bandwidth. The following table shows the different cables available and the bandwidth supported by them.

Cat5/5e is the most commonly found cable and offers approximately 1 Gbit/s bandwidth, however, given the choice go for Cat6 as it is pretty much priced the same and offers higher bandwidth. If your home has a network storage with multiple users accessing it at the same time then a Cat5/5e



will be sufficient but if you can go for Cat6 then why not. All the above cables are limited to a distance of 100 meters after which it is necessary to attach a repeater to regenerate the signal.